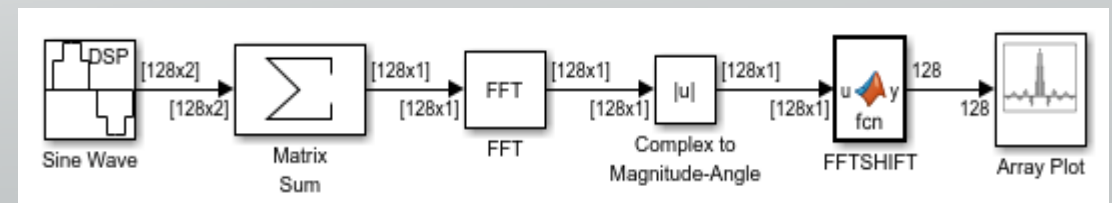
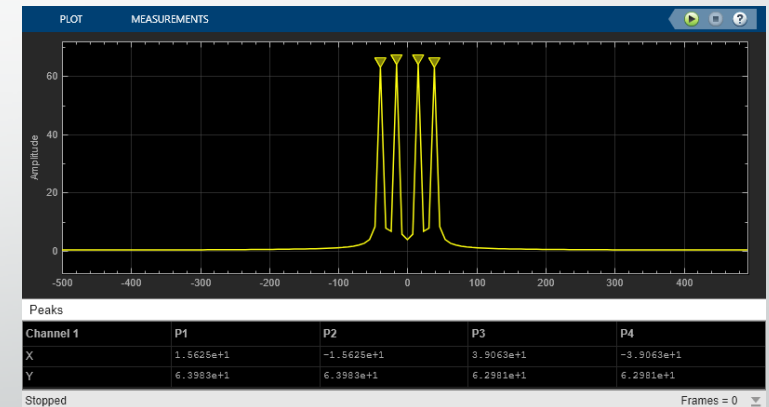


Fast Fourier Transform(FFT) Using Matlab Simulink



Quick review

$$X(t) = \sin(2\pi 15t) + \sin(2\pi 40t)$$

- For the example, we generated a sampling frequency and time vector and signal and then calculated FFT as blow:

% Compute Fourier transform of f(t)

clear

clc

Fs = 500; % Sampling frequency

T = 1/Fs; % Sampling period

L = 1500; % Length of signal

t = (0:L-1)*T; % Time vector

% Form a signal containing a 15 Hz sinusoid of
amplitude 1 and a 40 Hz sinusoid of amplitude 1.

X = sin(2*pi*15*t) + sin(2*pi*40*t);

% Plot the noisy signal in the time domain.

figure

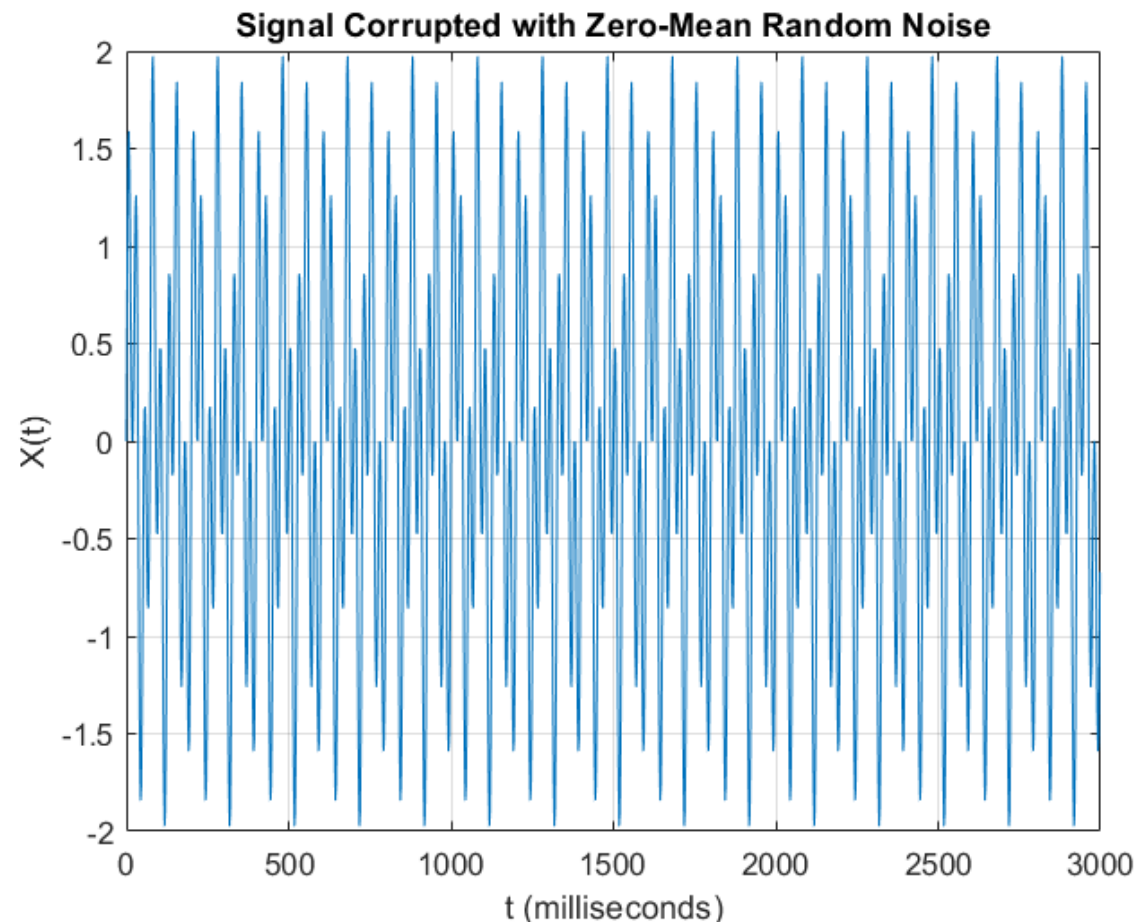
plot(1000*t,X)

title('Signal Corrupted with Zero-Mean Random Noise')

xlabel('t (milliseconds)')

ylabel('X(t)')

grid on



Quick review

$$X(t) = \sin(2\pi 15t) + \sin(2\pi 40t)$$

% Compute the Fourier transform of the signal.

Y = fft(X);

% Compute the two-sided spectrum P2. Then compute the single-sided spectrum P1 based on P2 and the even-valued signal length L.

P2 = abs(Y/L);

P1 = P2(1:L/2+1);

P1(2:end) = 2*P1(2:end);

% Define the frequency domain f1 and plot the single-sided amplitude spectrum P1.

f1 = (0:(L/2))*(Fs/L); % In sampled signal frequency repeats spectra at FS/2

figure

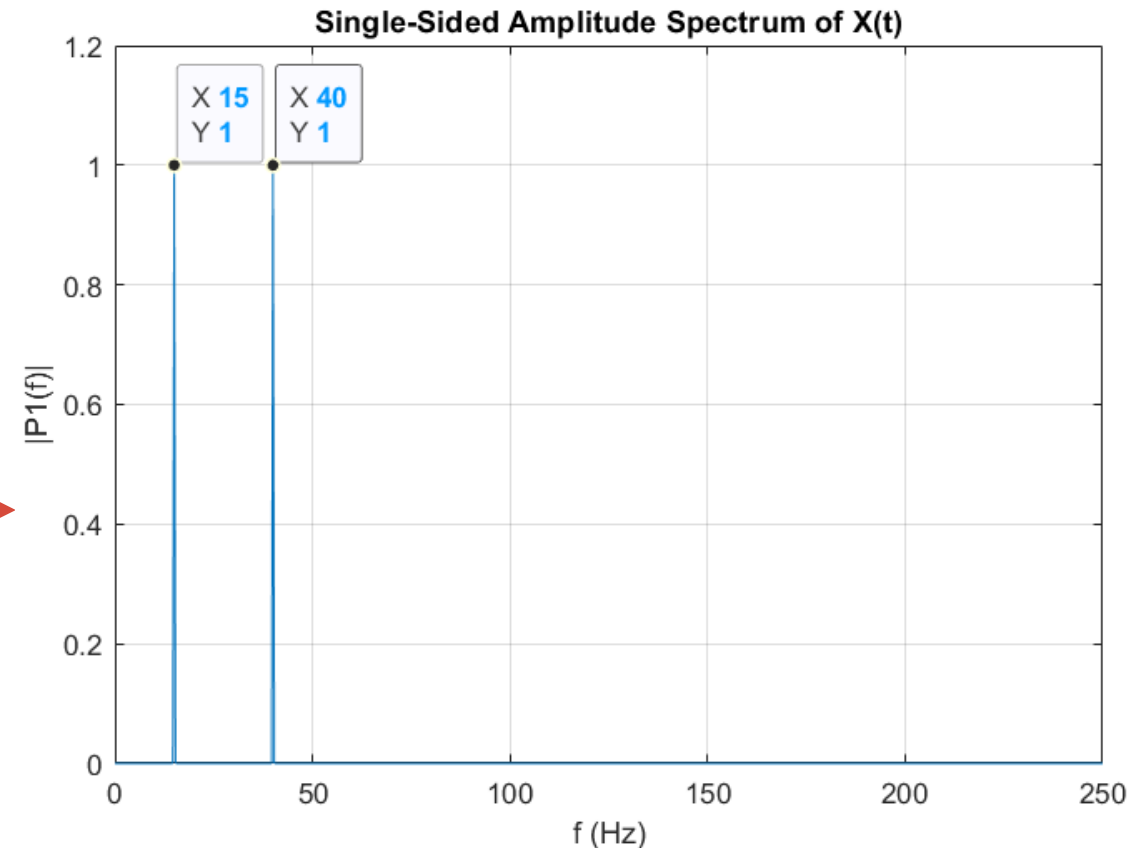
plot(f1,P1)

title('Single-Sided Amplitude Spectrum of X(t)')

xlabel('f (Hz)')

ylabel('|P1(f)|')

grid on



Quick review

% Define the frequency domain f2 and Shift the zero-frequency components then plot the Double-sided amplitude spectrum z.

```
f2 = (-L/2:L/2-1)*(Fs/L);
```

```
z = fftshift(Y);
```

```
figure
```

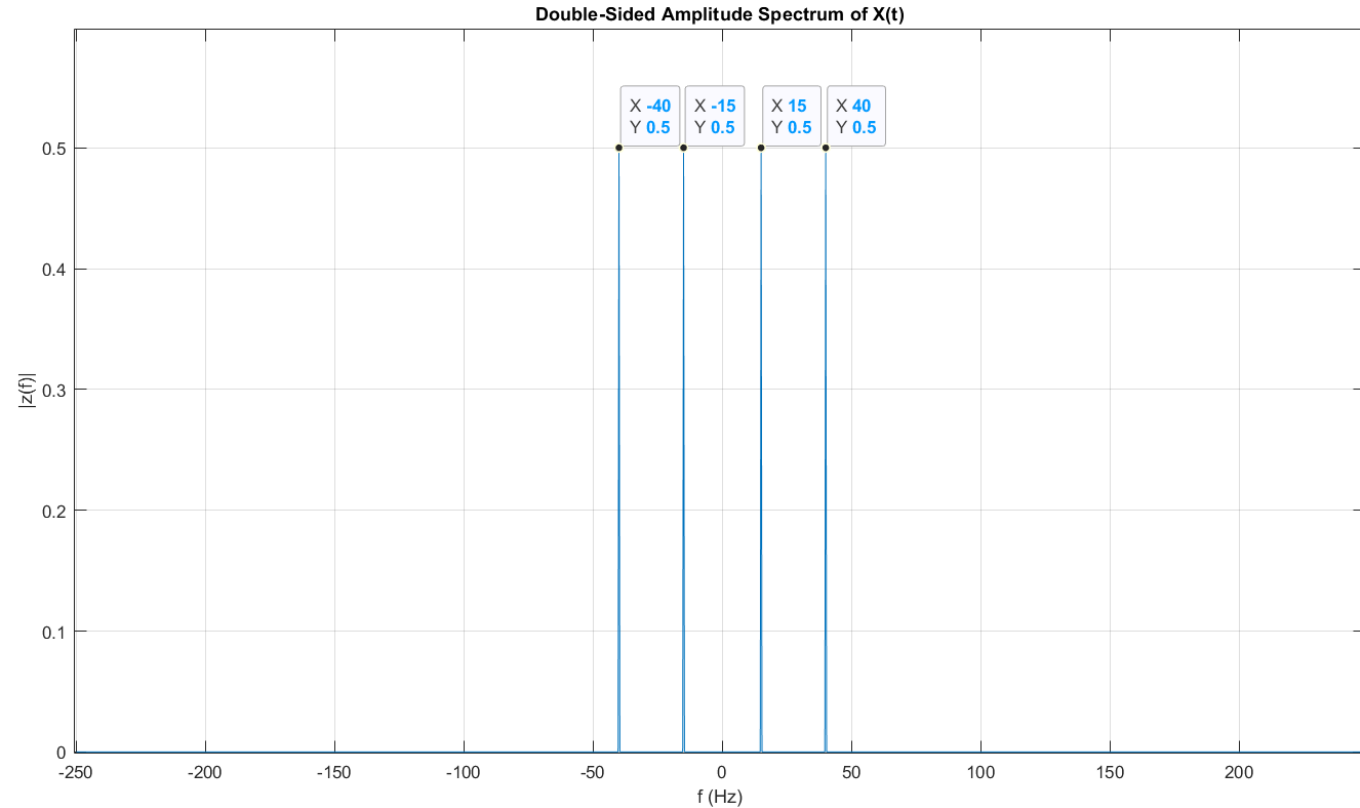
```
plot(f2,abs(z/L))
```

```
title('Double-Sided Amplitude Spectrum of X(t)')
```

```
xlabel('f (Hz)')
```

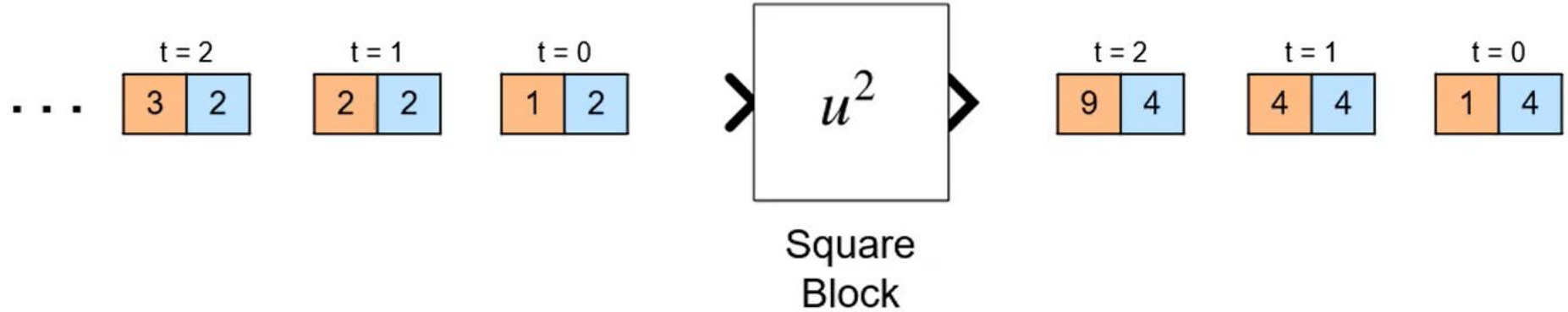
```
ylabel('|z(f)|')
```

```
grid on
```

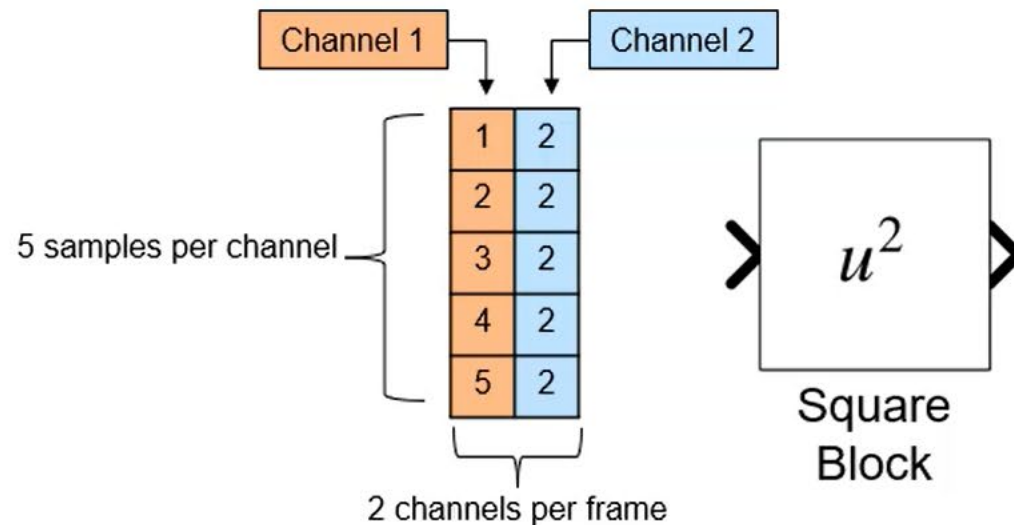


Sample-Based vs Frame-Based Processing

Sample-Based Processing



Frame-Based Processing



Sample-Based vs Frame-Based Processing

- Advantages of frame-based processing
 - Improved simulation performance
 - Emulate real-time signal processing systems and data acquisition hardware
- Simulink provides functionality for frame-based signal processing

FFT using Simulink

- Use the Sine Wave block to generate two sinusoids, one at 15 Hz and the other at 40 Hz and the sampling rate (1kHz). Use the Matrix Sum block to add the sinusoids
- point-by-point to generate the compound sinusoid:

$$u = \sin(2 * 15\pi t) + \sin(2 * 40\pi t)$$

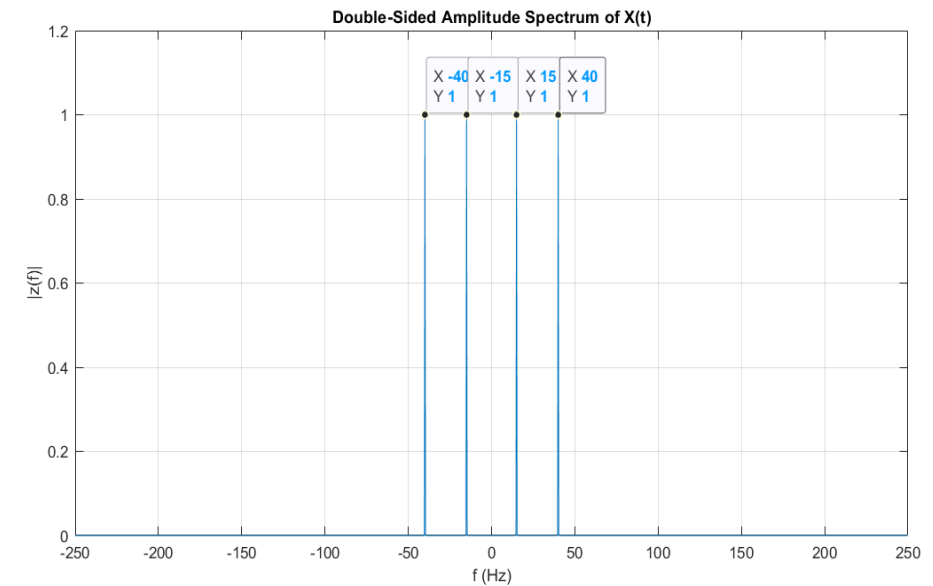
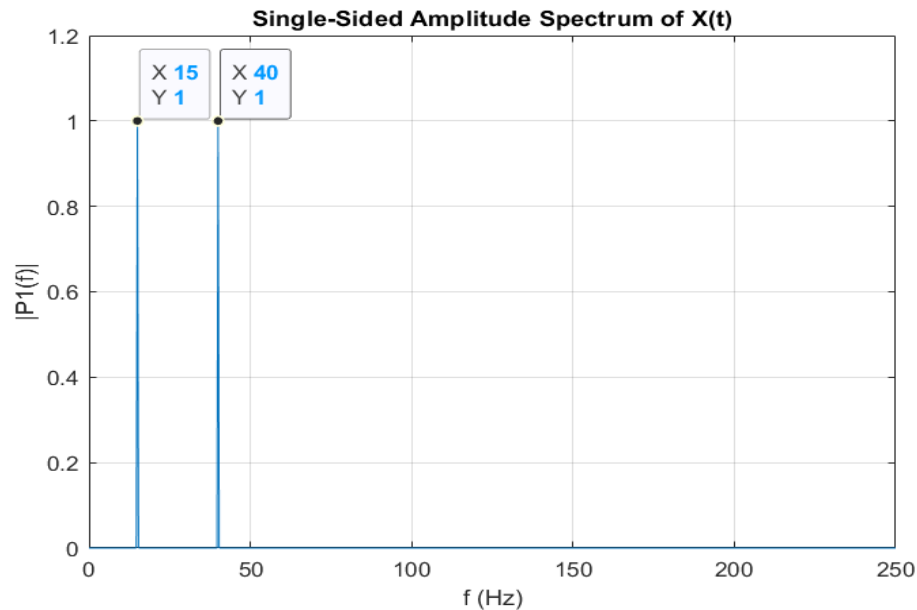
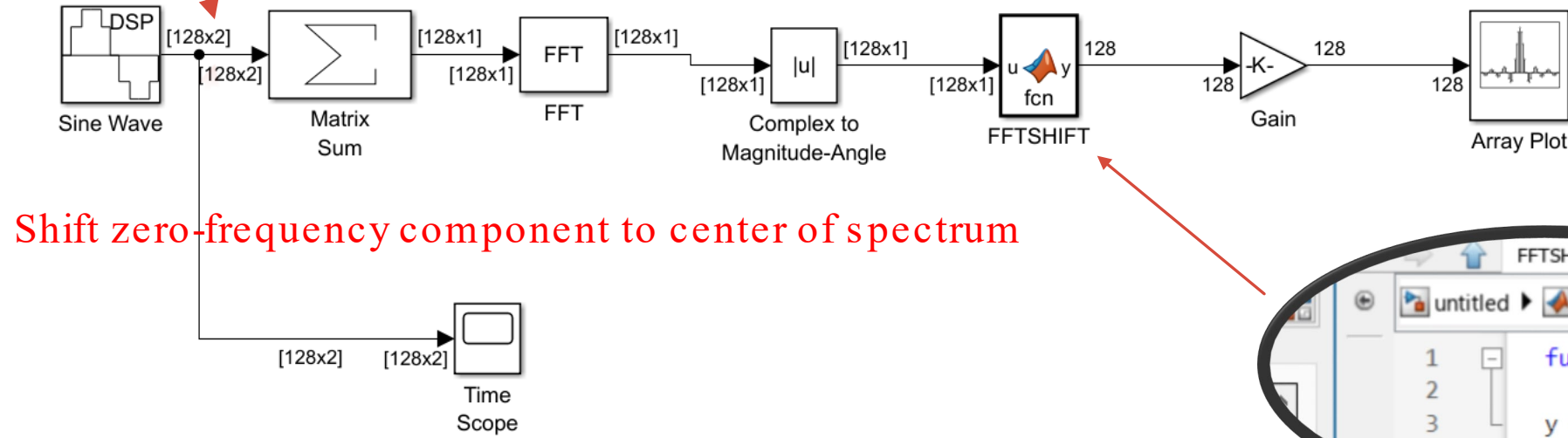
- Transform this sinusoid into the frequency domain using an FFT block.

??



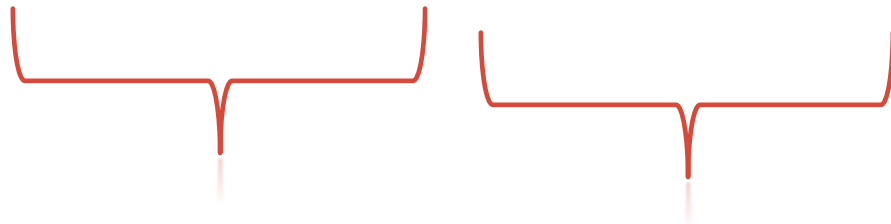
FFT using Simulink

??



FFT using Simulink

$$u = \sin(2 * 15\pi t) + \sin(2 * 40\pi t)$$



Block Parameters: Sine Wave

Sine Wave (mask) (link)

Output samples of a sinusoid. To generate more than one sinusoid simultaneously, enter a vector of values for the Amplitude, Frequency, and Phase offset parameters.

Main Data Types

Amplitude: 1

Frequency (Hz): [15 40] [15,40]

Phase offset (rad): 0

Sample mode: Discrete

Output complexity: Real

Computation method: Trigonometric fcn

Sample time: 0.001

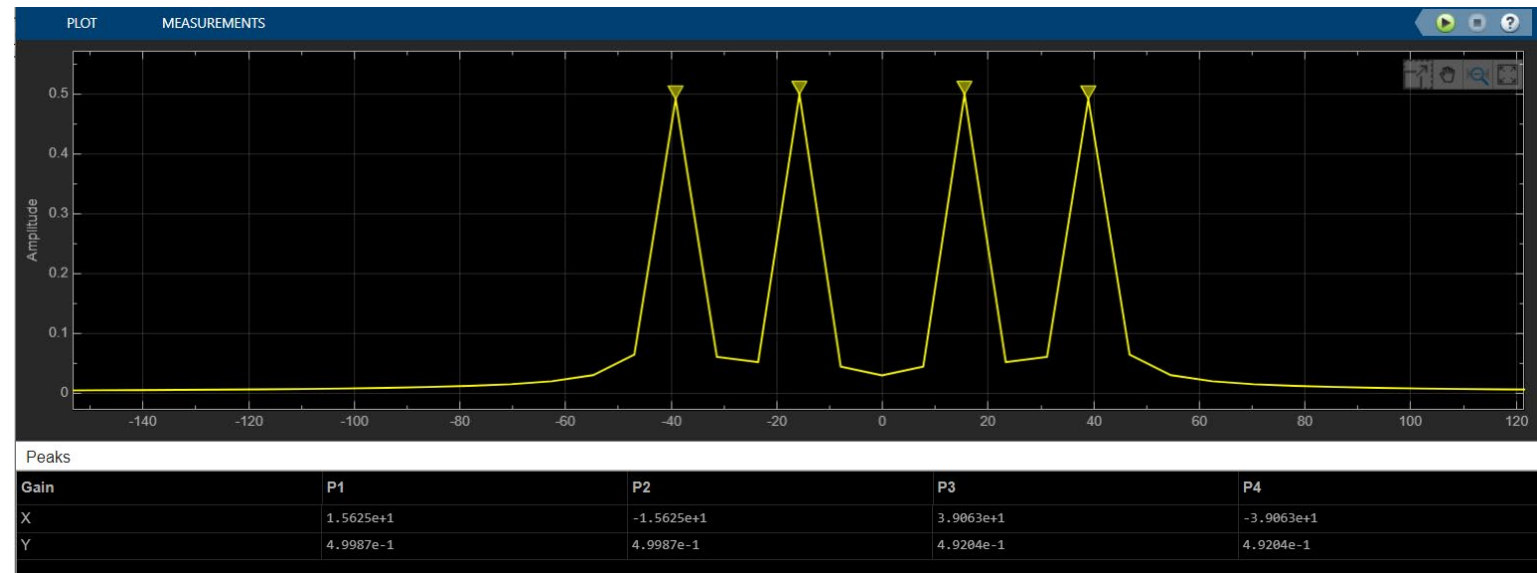
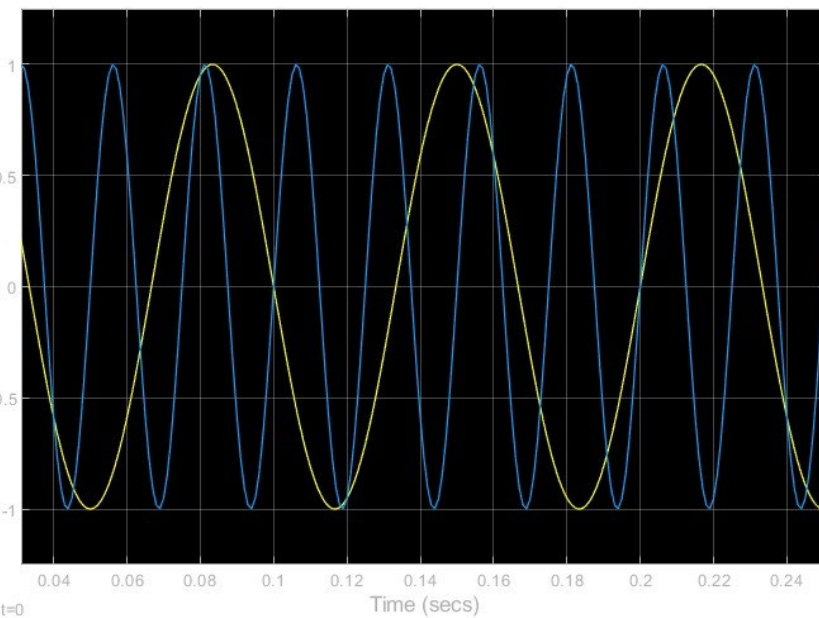
Samples per frame: 128

Resetting states when re-enabled: Restart at time zero

OK Cancel Help Apply

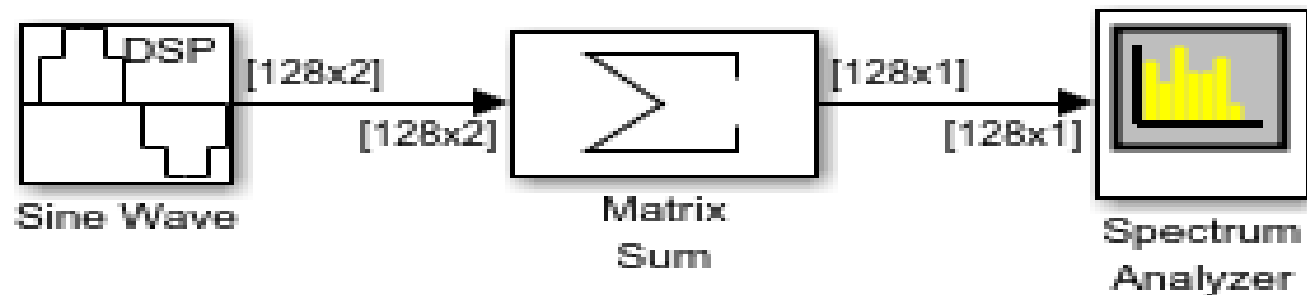
FFT using Simulink

- The scope shows peaks at 15 and 40 Hz, as expected. You have now transformed two sinusoidal signals from the time domain to the frequency domain.



FFT using Simulink

- You can use a Spectrum Analyzer block in place of the sequence of FFT, Complex to Magnitude-Angle, MATLAB Function, and Array Plot blocks. The Spectrum Analyzer computes the magnitude FFT and shifts the FFT internally.



FFT using Simulink

- The blocks in the Power Spectrum Estimation library compute the FFT internally.

